

Fuerza magnética: $\vec{F}_q = q\vec{v} \times \vec{B} \qquad d\vec{F}_I = I d\vec{\ell} \times \vec{B}.$

Torque magnético: $\vec{\tau} = \vec{r} \times \vec{F} \qquad d\vec{\tau} = \vec{r} \times d\vec{F}.$

$$\vec{\mu} = I\vec{A}.$$

$$\vec{\tau} = \vec{\mu} \times \vec{B}.$$

Biot-Savart: $\vec{B} = \frac{\mu_0}{4\pi} q \frac{\vec{v} \times \hat{r}}{r^2} \qquad d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{\ell} \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}.$

Ley de Ampère: $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}}.$

Flujo Magnético: $\Phi_B = \int \vec{B} \cdot d\vec{A}.$

Ley de Faraday-Lenz: $\varepsilon_{\text{ind}} = -\frac{d\Phi_B}{dt}.$

$$\varepsilon_{\text{ind}} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{\ell}.$$

Diferencia de potencial: $\Delta V = -\int \vec{E} \cdot d\vec{\ell}.$

I) Integrales:

$$\int \frac{du}{(u^2 + 1)^{3/2}} = \frac{u}{\sqrt{u^2 + 1}}.$$

$$\int \frac{u du}{(u^2 + 1)^{3/2}} = -\frac{1}{\sqrt{u^2 + 1}}.$$